Programming Concepts -- Guided Notes ANSWER KEY

Unit tp-3.9 | CompTIA Network+ N10-009 Obj. FC0-U71 3.9 | N10-008 FC0-U71 3.9

For instructor use / distribute after students complete the notes.

Question 1

A _______ stores a single value that can change during program execution. A _______ stores a value that does not change. A _______ is a named location that holds one value at a time. A _______ (like a list/array) can hold multiple values under one name.

Answer: Variable; constant; variable; data structure (array/list).

Question 2

Data types define what kind of data a variable holds. Common types: _______ (whole numbers), _______ (decimal numbers), _______ (text), _______ (True/False). Assigning a value of the wrong type to a variable can cause a _______ error.

Answer: Integer (int); float (floating-point); string (str); boolean (bool); type (runtime).

Question 3

OOP (Object-Oriented Programming) models software around _______ that combine data (_______) and behavior (_______). A _______ is a blueprint; an _______ is a specific object created from that blueprint. OOP supports _______ (hiding internal details) and _______ (child classes inherit from parent).

Answer: Objects; attributes (properties); methods (functions); class; object (instance); encapsulation; inheritance.

Question 4

A syntax error is a violation of the _______ rules of a programming language and is caught at _______ time. A logic error is a bug where the code runs without crashing but produces the _______ output. A runtime error occurs while the program _______ and causes it to _______ unexpectedly.

Answer: Grammar (syntax); compile/parse; wrong; runs; crash.

Question 5

Version control (like Git) tracks _______ to source code over time. A _______ is a saved version of the codebase. _______ allows developers to create an isolated copy of the code to work on a feature without affecting the main codebase. _______ combines changes from two branches.

Answer: Changes (edits); commit; branching; merging.

Question 6

An API (Application Programming Interface) defines how two software components can _______ with each other. When a weather app on your phone displays today's forecast, it is likely calling a weather service _______ to get the data. APIs use _______ requests over HTTP to send and receive _______ (typically JSON or XML format) data.

Answer: Communicate; API; HTTP (GET/POST); structured.

Question 7

Testing ensures software functions as expected. _______ testing checks individual functions in isolation. _______ testing checks how multiple components work together. _______ testing verifies the complete system meets requirements from the user's perspective. _______ testing ensures that new changes do not break existing functionality.

Answer: Unit; integration; end-to-end (system/user acceptance); regression.

Question 8 -- Real World Application

REAL WORLD: A student writes a program to calculate the area of a rectangle. They test it and it runs without errors, but the area value it prints is always wrong. Identify what type of error this is, explain why it does not show up as a crash, and describe how the student should debug it.

Answer: This is a logic error -- the code is syntactically correct (no grammar violations) and does not crash (no runtime exception), but it produces incorrect output. Logic errors are the most difficult to catch because the program appears to work. Debugging approach: (1) Print intermediate values at each step to trace what the program is actually calculating. For example, print the length and width values before the calculation to verify they are being read correctly. (2) Manually work through the formula and compare to the program output -- check the formula itself (is area = length x width, or did they accidentally use length + width?). (3) Check operator precedence if complex expressions are involved. (4) Use an IDE debugger to step through the code line by line and inspect variable values at each step.